Algorithmic Trading Strategy

UMN FMA Trading Competition

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What We Built

Two autonomous strategies trading US equities & ETFs

Strategy	**Allocation**	**Asset(s)**
LSTM Portfolio	85%	VTI, SCHZ, PDBC, VIXM
RSI-2 Mean Reversion	15%	QQQ

All trades execute daily before market close via Alpaca API

What is an LSTM?

LSTM (Long Short-Term Memory) is a type of recurrent neural network designed to learn patterns in sequential data.

- Standard neural networks treat each input independently
- LSTMs have **memory**: they remember relevant information from previous timesteps and forget what is no longer useful
- This makes them well-suited for **time series** data like stock prices, where the order and history of observations matter

In our case: the LSTM reads 50 days of market data sequentially, builds an internal representation of current market conditions, and predicts how to allocate a portfolio tomorrow.

Reference: Hochreiter & Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.
Adapted from: Belmudes, "Quant Finance Projects,"
`github.com/BelmudesLSB/quant-finance-projects`

Strategy 1: LSTM Portfolio (85%)

A neural network decides how to split money across 4 asset classes **every day**

- **VTI**: US Stocks
- **SCHZ**: Bonds
- **PDBC**: Commodities
- **VIXM**: Volatility

Phase	Period	Purpose
Training	Jan 2016 to Feb 2025	Model learns patterns
Validation	Mar 2025 to Feb 2026	Model selection, prevents overfitting

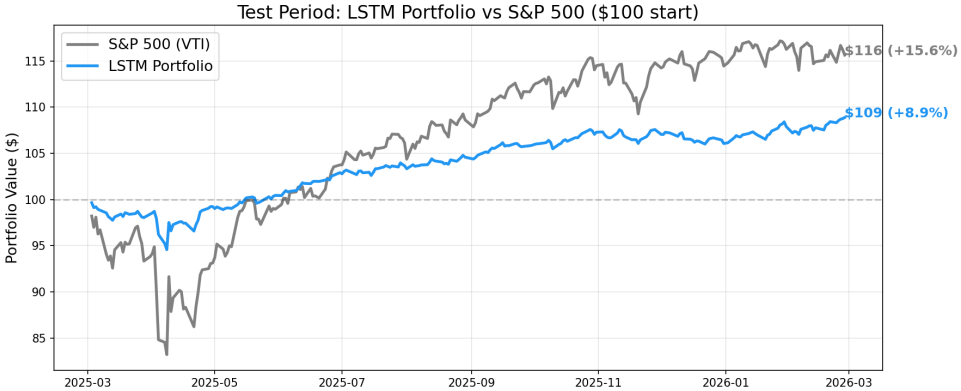
Optimized for **Sharpe ratio**: maximizes returns *per unit of risk*

How the LSTM Works

1. Feeds in last **50 days** of prices & returns for all 4 assets
2. LSTM processes the sequence, learns patterns
3. Outputs **4 weights** (e.g., 28% stocks, 13% bonds, 30% commodities, 29% volatility)
4. Rebalances portfolio to match

Key: In uncertain markets, the model *automatically* shifts toward bonds, commodities, and volatility

LSTM Backtest: Validation Period (Mar 2025 to Feb 2026)



S&P 500: +15.6% return
Max drawdown: −16%

Our LSTM: +8.9% return
Max drawdown: **only −5%**

Max drawdown = largest peak-to-trough decline during the period. Lower is better.

Strategy 2: RSI-2 Mean Reversion on QQQ (15%)

RSI (Relative Strength Index) measures how overbought or oversold an asset is

We use a 2-day RSI, computed from the last 2 daily price changes:

$$\text{RSI}(2) = 100 - \frac{100}{1 + \dfrac{\text{average gain over 2 days}}{\text{average loss over 2 days}}}$$

- RSI near 0 = extremely oversold (price dropped sharply)
- RSI near 100 = extremely overbought (price surged)

Our rules:

- **Buy QQQ** when $\text{RSI}(2) < 10$: sharp drop, expect a bounce
- **Sell QQQ** when $\text{RSI}(2) > 70$: let profits run, then take profit
- Typical hold: 1–5 days

RSI-2 Backtest: Performance During Market Crises

Period	**Dates**	**S&P 500**	**QQQ**	**RSI-2**
Russia-Ukraine War	01-Feb-22 to 30-Jun-22	−16.1%	−22.8%	**+2.0%**
2022 Bear Market	01-Jan-22 to 31-Dec-22	−19.5%	−33.0%	**+3.3%**
2025 Tariff Crisis	01-Mar-25 to 30-May-25	−0.8%	+2.2%	**+13.4%**
Recent 3 Months	01-Dec-25 to 27-Feb-26	+0.4%	−1.9%	**+1.9%**

Risk Management

1. **Diversified**: 4 asset classes, not just stocks
2. **AI-driven defense**: LSTM shifts to safe havens automatically
3. **Position sizing**: 85/15 split limits RSI exposure
4. **Long-only**: no leverage, no shorts, no options
5. **Daily rebalancing**: never holds stale positions
6. **All orders before close**: fills at real-time prices

Execution

Infrastructure

- Linux server running 24/7
- Python + PyTorch + Alpaca API
- 2 systemd services with auto-restart on failure

Daily Schedule (ET)

- 3:45 PM: LSTM fetches data, runs model, rebalances 4 ETFs
- 3:50 PM: RSI-2 checks QQQ signal, trades if triggered
- All orders fill before 4:00 PM close
- Max 4–5 orders per day

Summary

1. **LSTM (85%)**: AI allocates across stocks, bonds, commodities, volatility
2. **RSI-2 (15%)**: Buys QQQ dips, sells after recovery

Designed for uncertain markets: protects capital while capturing upside

github.com/verma-bipul/Algo-Trade-UMN-Team4